ERHVERVSAKADEMI
AARHUS

Machine Learning – Day 4

Sila, Sept 16th 2026.

Welcome - Todays Program.

- Comments about last weeks exercises.
- Support Vector Machines (SVM),
- Clustering, Decision Trees, Random Forest and Ensemble Methods.
- Exercises

Machine Learning – The Iris DataSet



Last week.

We looked at the Iris Dataset

https://archive.ics.uci.edu/ml/datasets/Iris

Exercise 3.5. – from last week.

In groups of 2-3.

Carefully, together, go through last weeks exercise 3.5
+ the followup exercise ”exercise iris precision”:

Discuss:
- How did you solve the exercise?
- Problems understanding the code?
- Comments?

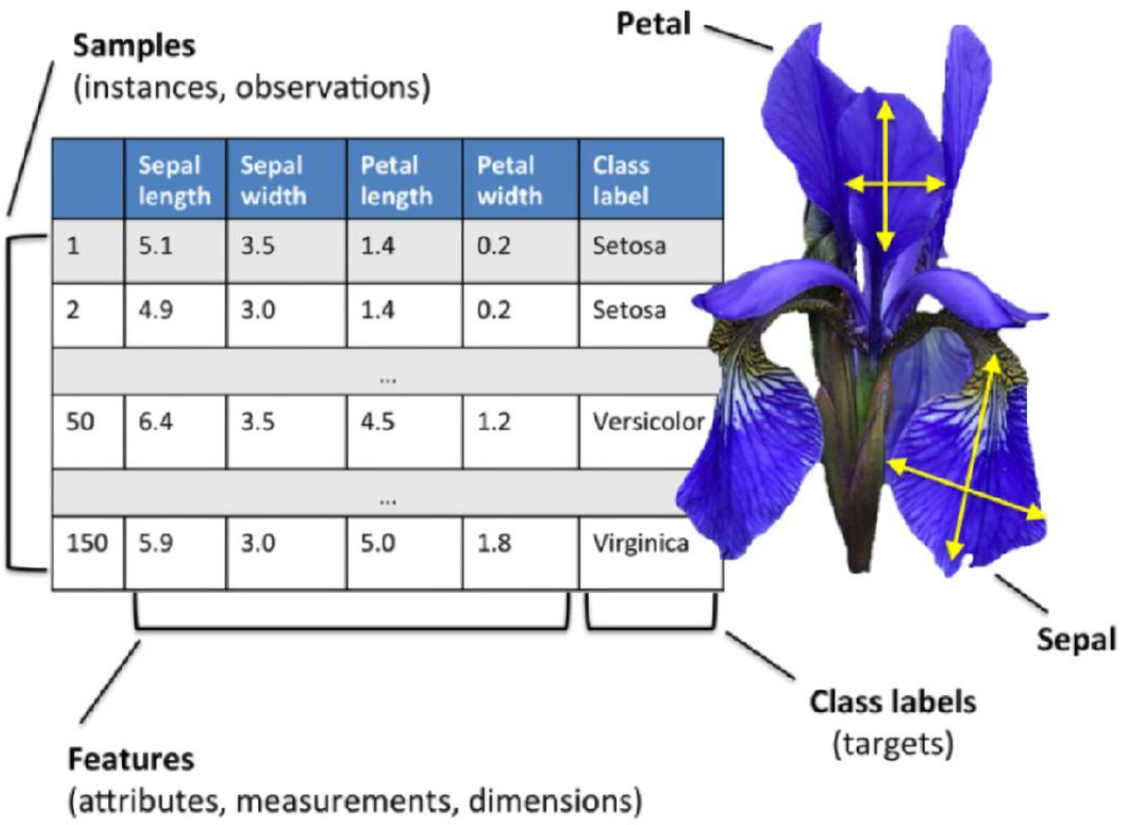
Logictic regression and the Iris data set:
Exercise 3.5 and the exercise
"exercise_iris_precision" from
last week (with Logistic regression exercises)

Exercises – More about the Iris Dataset.

In groups of 2-3.

Lets look at some more (logistic regression) code for the Iris data set.
Explain to each other what the code is doing.

See:
Exercise_Iris_Precision.py on Canvas.



	Sepal length	Sepal width	Petal length	Petal width	Class label
1	5.1	3.5	1.4	0.2	Setosa
2	4.9	3.0	1.4	0.2	Setosa
...					
50	6.4	3.5	4.5	1.2	Versicolor
...					
150	5.9	3.0	5.0	1.8	Virginica

Machine Learning – Support Vector Machines

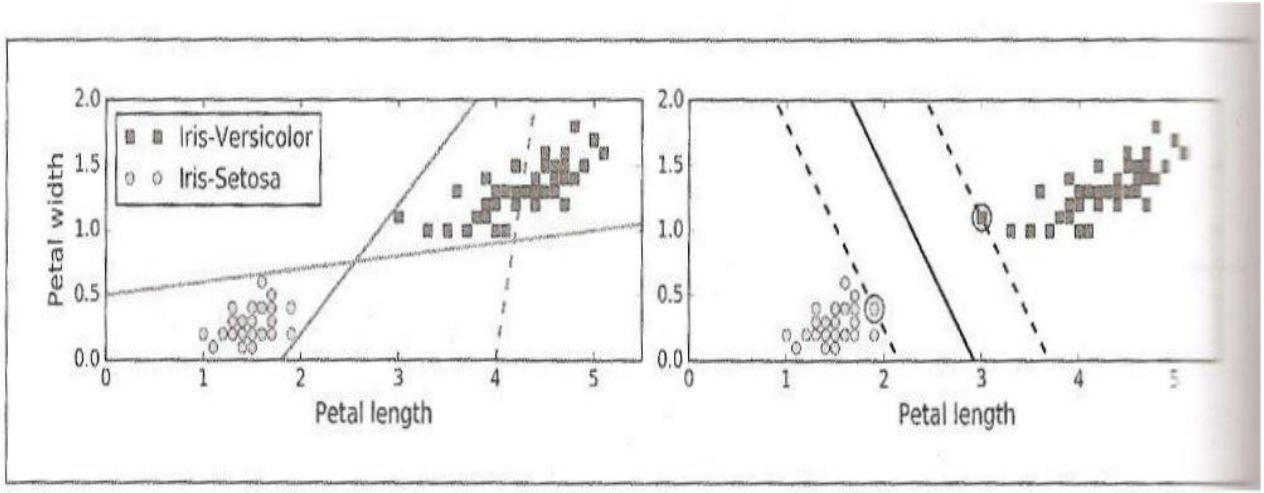
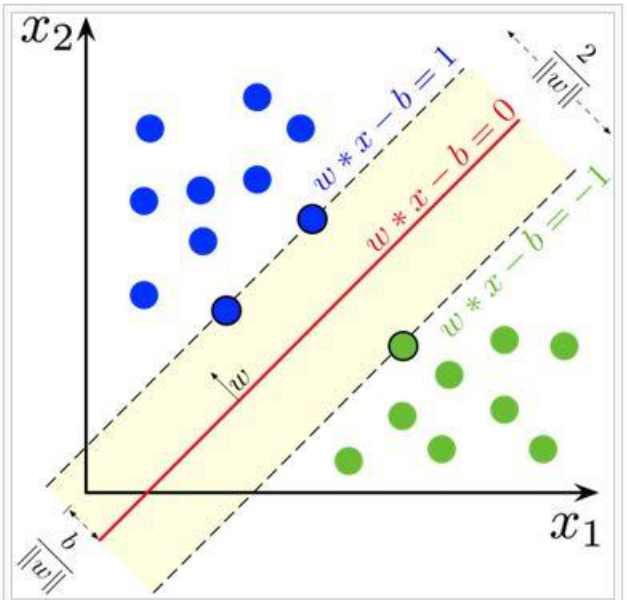


Figure 5-1. Large margin classification

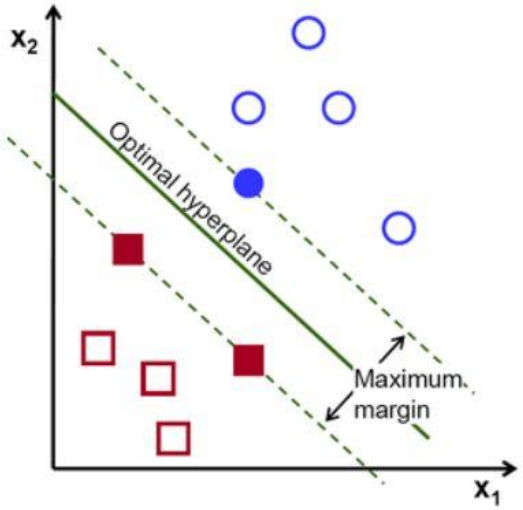
You can think of an SVM classifier as fitting the widest possible street (between the classes). This is calles "large margin classification".

p. 154 in "Hands on" by Geron.

Machine Learning – Support Vector Machines



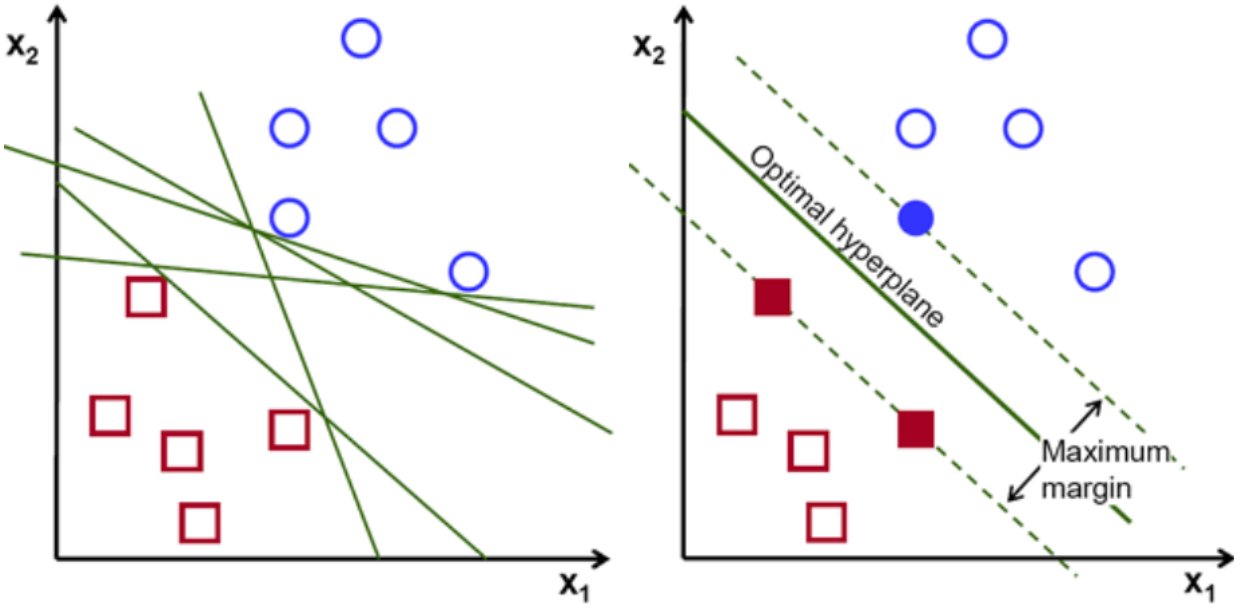
Maximum-margin hyperplane and margins for an SVM trained with samples from two classes. Samples on the margin are called the support vectors.



You can think of an SVM classifier as fitting the widest possible street (between the classes). This is calles "large margin classification".

p. 154 in "Hands on" by Geron.

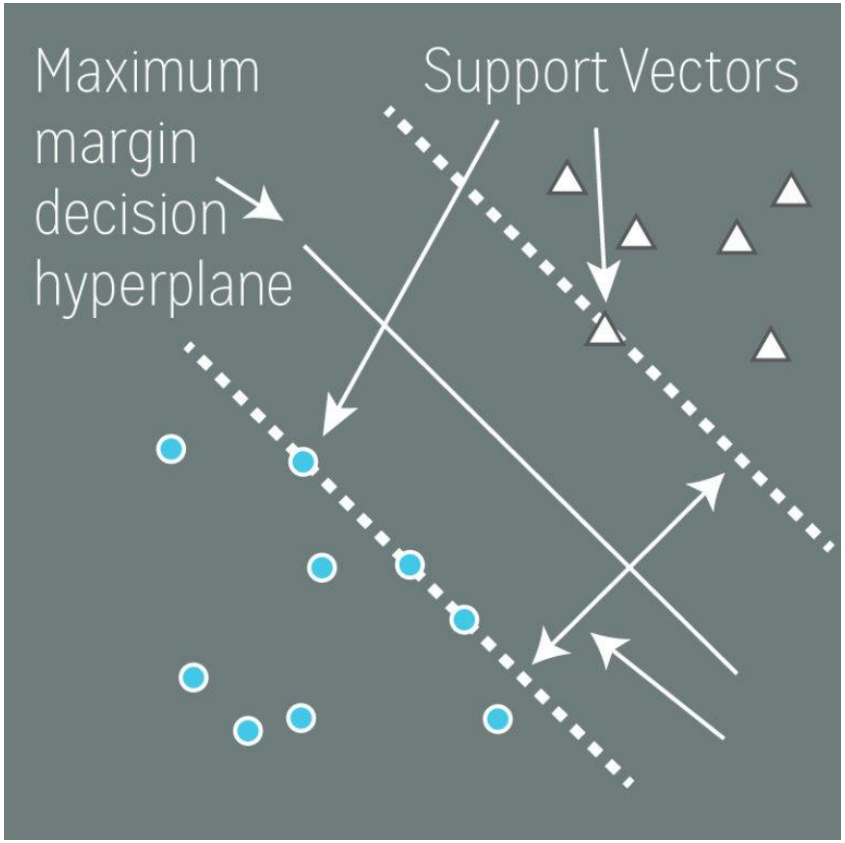
Machine Learning – Support Vector Machines



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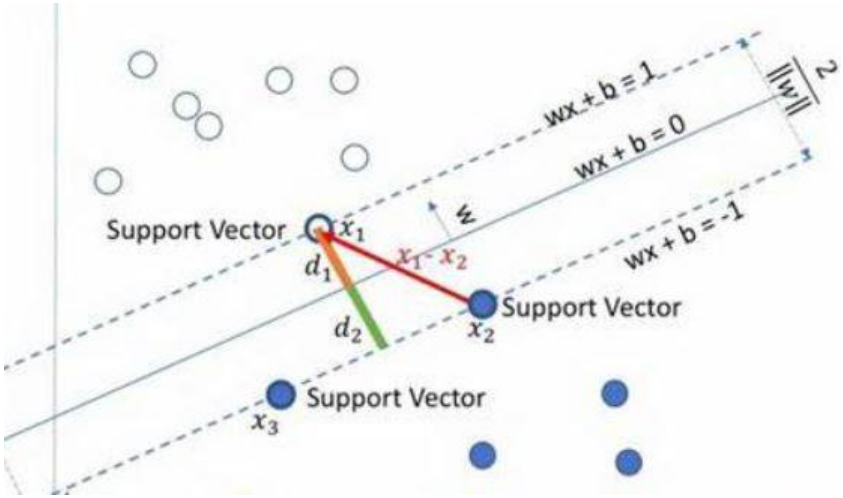
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Machine Learning – Support Vector Machines



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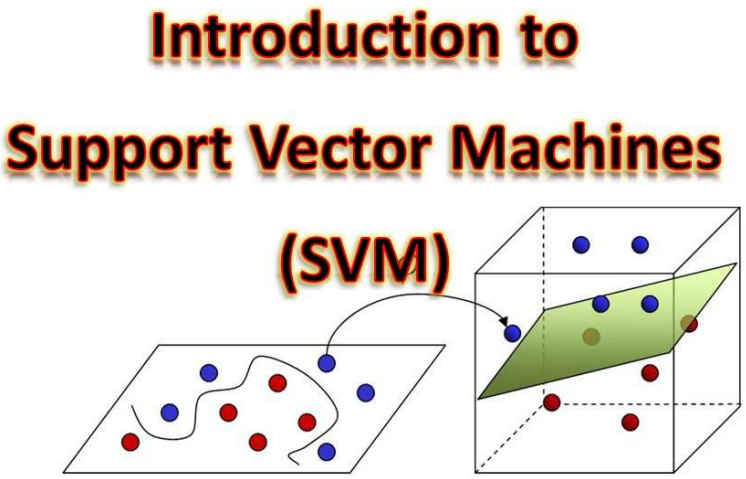
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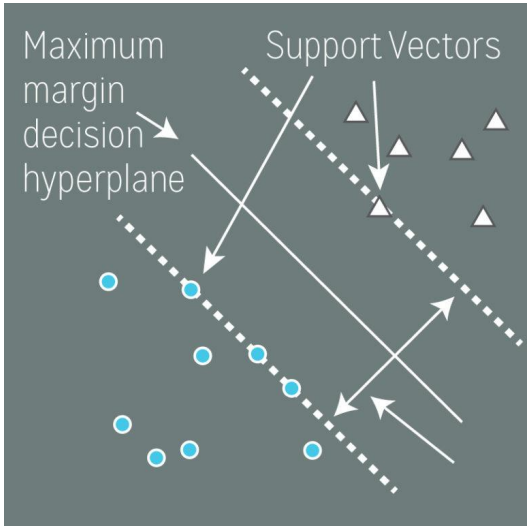


Machine Learning – Support Vector Machines

You can think of an SVM classifier as fitting the widest possible street (between the classes). This is calles "large margin classification".

p. 154 in "Hands on" by Geron.





Machine Learning – Support Vector Machines

Use SVM software package (e.g. liblinear, libsvm, ...) to solve for parameters θ.

Need to specify:

→ Choice of parameter C.

Choice of kernel (similarity function):

E.g. No kernel ("linear kernel")

Predict "y = 1" if $\theta^T x \geq 0$

$\theta_0 + \theta_1 x_1 + \cdots + \theta_n x_n \geq 0$

→ n large, m small

$x \in \mathbb{R}^{n+1}$

Gaussian kernel:

$$f_i = \exp\left(-\frac{||x - l^{(i)}||^2}{2\sigma^2}\right), \text{ where } l^{(i)} = x^{(i)}.$$

Need to choose σ^2.

$x \in \mathbb{R}^n$, n small and/or m large



Andrew Ng, Coursera

Machine Learning – Support Vector Machines

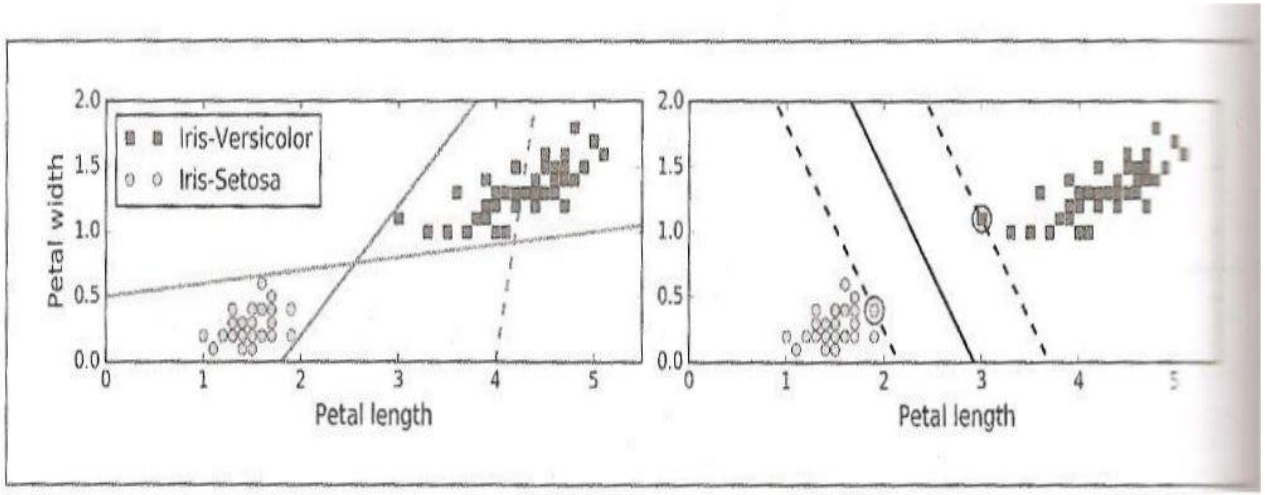


Figure 5-1. Large margin classification

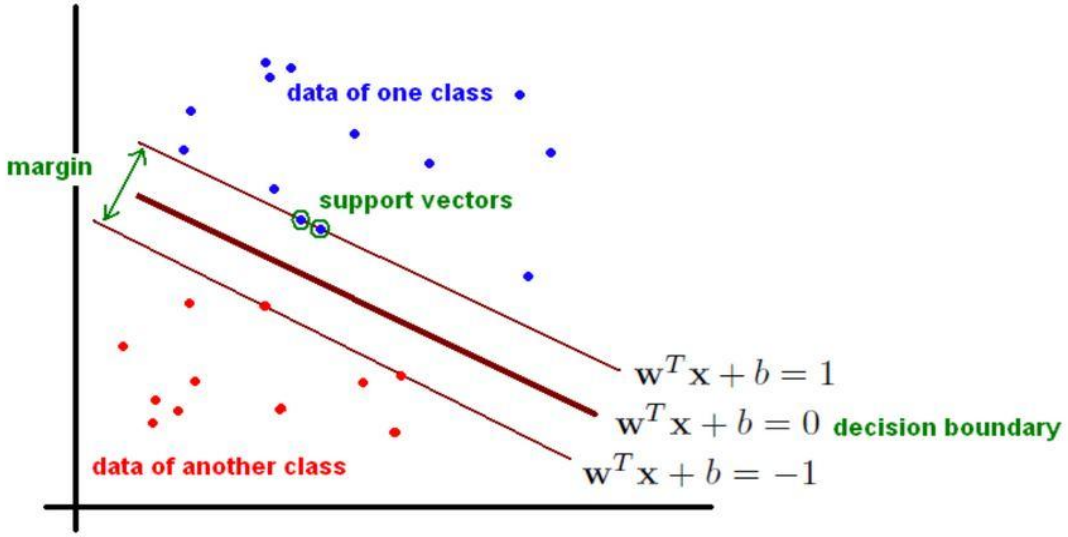
Many datasets are not linearly separable.
SVMs can handle some cases of this by adding ”kernels”.
We will look at examples today, but won't go into the math.

See p. 154 in "Hands on" by Geron.

Machine Learning – Support Vector Machines

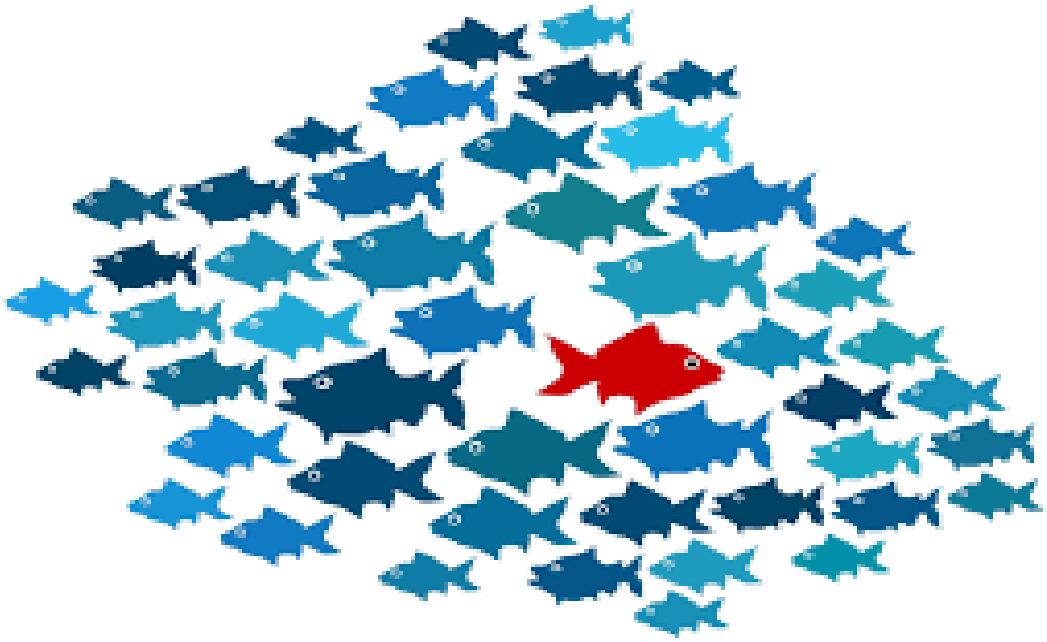
Support Vector Machines

Exercises… Code: Svm1, Svm2 & Svm_gamma



Recap: What can Machine Learning do?

Detect anomalies (security, fraud).



Recap: What can Machine Learning do?

Healthcare – Second opinion:

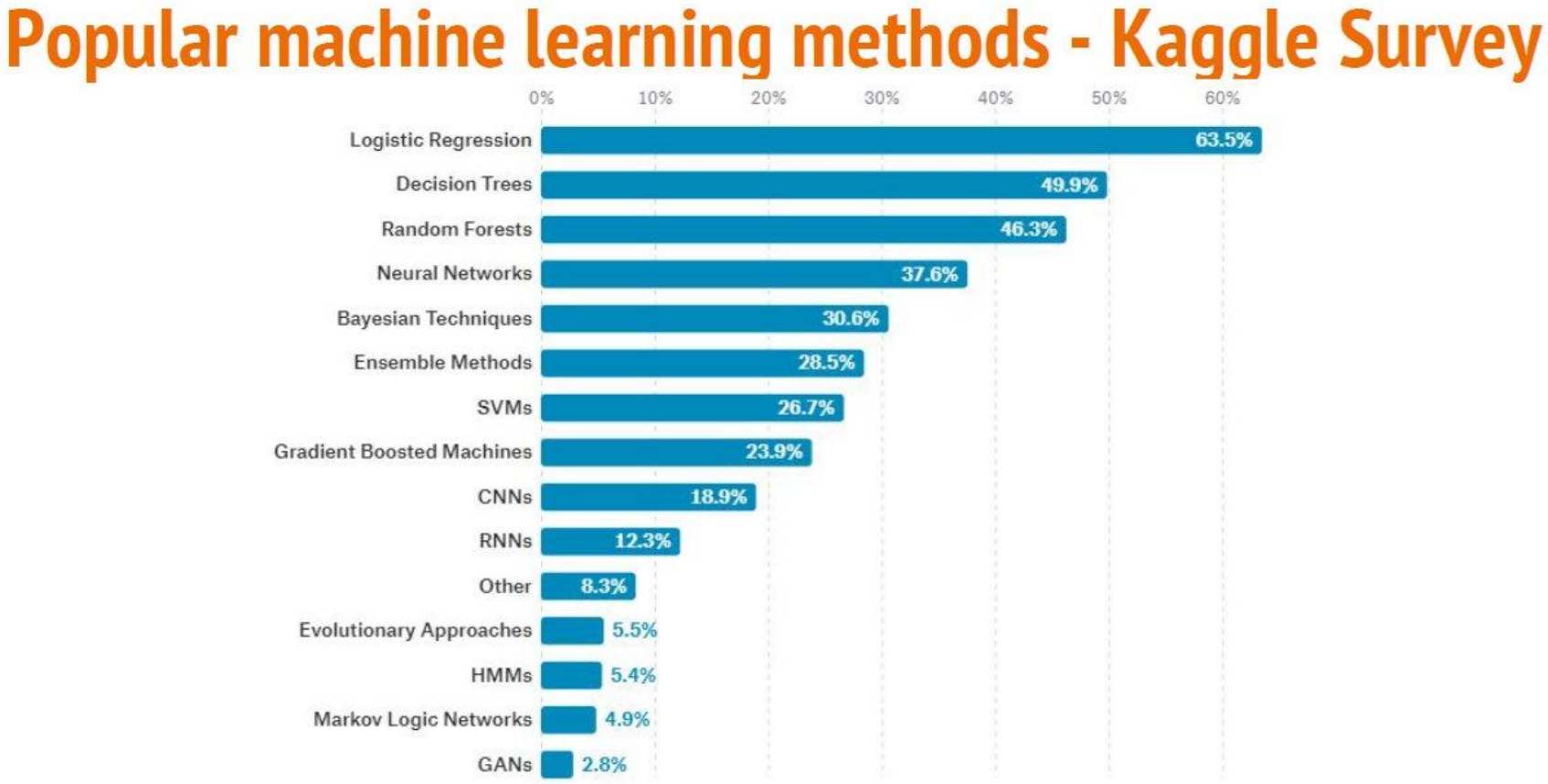
- Historical datasets of patient characteristics.
- Data and algorithm are combined to create predictive model.
- Predictive models assigns probabilities.
- Patients outcome is predicted.

Recap: What can Machine Learning do?

Customer segmentation
for targeted marketing.



Machine Learning – Methods.



Machine Learning – Two general problems.

Two general problems:

1. We have some data with labels (categories) and we would like to assign a category to new data with no labels.

2. We have some data without labels, which we would like to divide into groups and then use those groups for new data.

Number 1 is called supervised learning (many techniques exist for this).
Number 2 is called clustering (one example of unsupervised learning).

Machine Learning – Clustering, Decision Trees, Random Forest, Ensemble methods.

Today:

- Clustering,
- Decision Trees,
- Random Forest

and

- Ensemble Methods...

Clustering

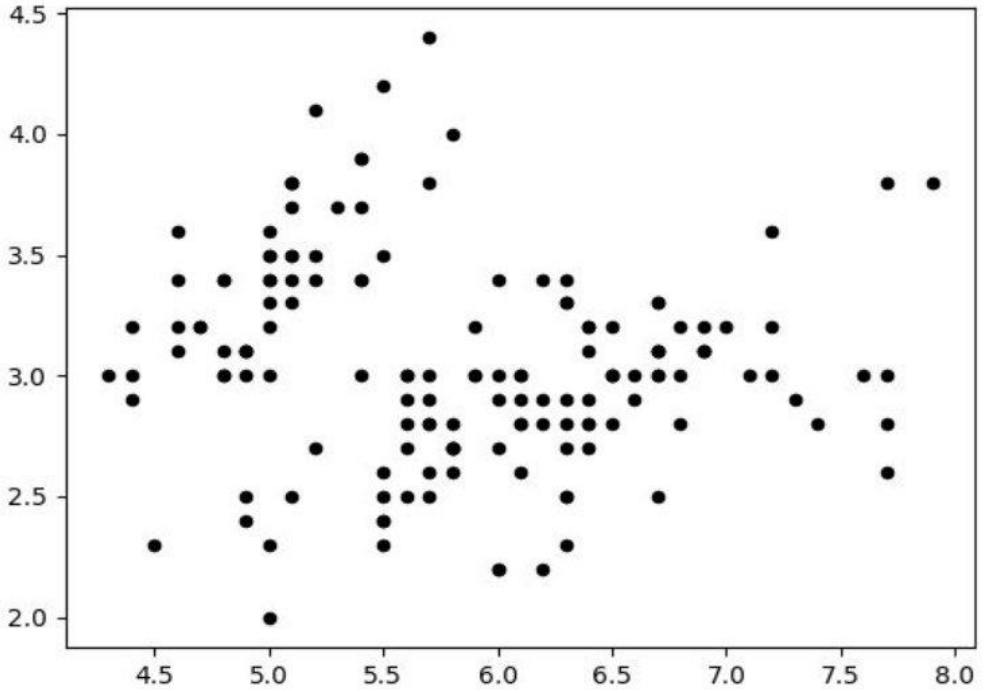
Clustering : The Problem (non-labelled data)

We have some data that we would like to group in a smart way.

- For instance, we would like to group our customers from a webshop into categories based on their behaviour.
- Typically data consists of multidimensional attributes. For instance a customer could have attributes such as : age, zipcode, average spending per month, country, gender (or more). Here we have 5 dimensions.
- The problem is then how can we group our input data, such that the "distance" inside the groups are small (similar) and the distance between the groups are big (so we get distinct groups).
- A related question is: How many groups do we need?

Machine Learning – Clustering.

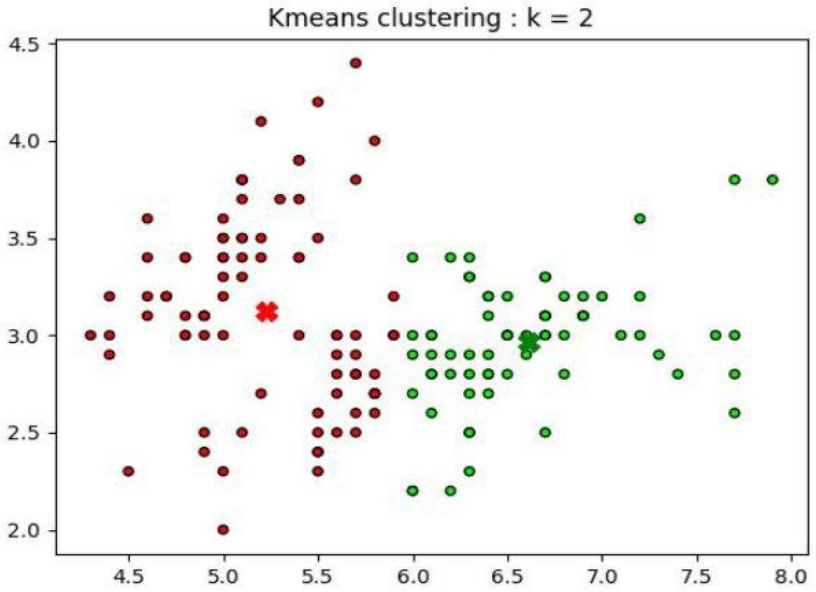
Clustering - an example - 1 cluster

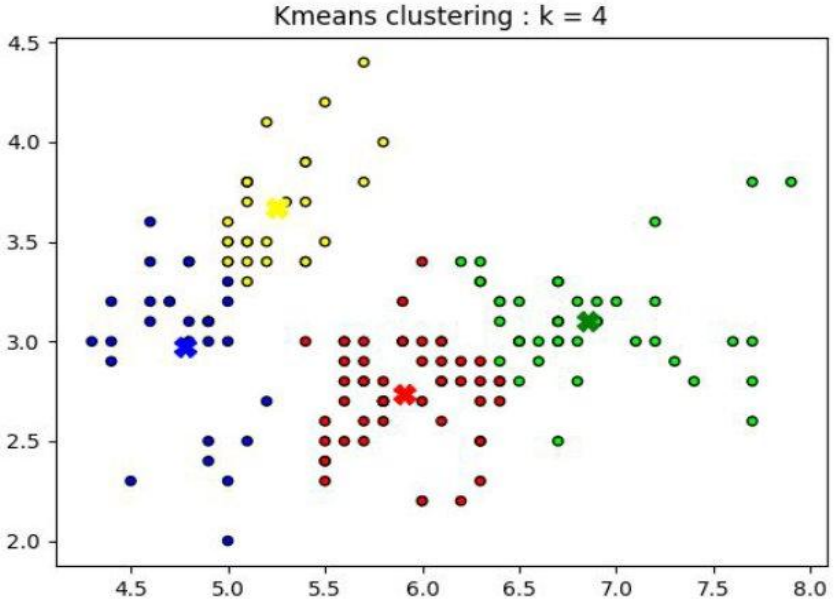


How many dimensions does this data have?

Machine Learning – Clustering.

2 clusters vs 4 clusters - notice cluster centers





Clustering - k-means algorithm

Input: some multidimensional data (unlabeled) and k (number of clusters).

The algorithm will build k numbers of clusters while obeying this rule:
1. Each point in a cluster is closer to the cluster center than to other cluster centers. [distance calculated using multi-dimensional pythagoras]

A cluster center is the arithmetic mean (average) of all the points in the cluster.

Finding optimal clusters is NP-hard (meaning unless you try ALL possible clusters - exponential growth) you cannot be guaranteed to find the optimal clusters

K-means - Scikit Learn implementation

The k-means problem is solved using Lloyd's algorithm. (https://en.wikipedia.org/wiki/Lloyd%27s_algorithm)

Iterative algorithm, guaranteed to converge, but can land in a suboptimal Solution.

It basically switches between two operations (starts with k random centers):

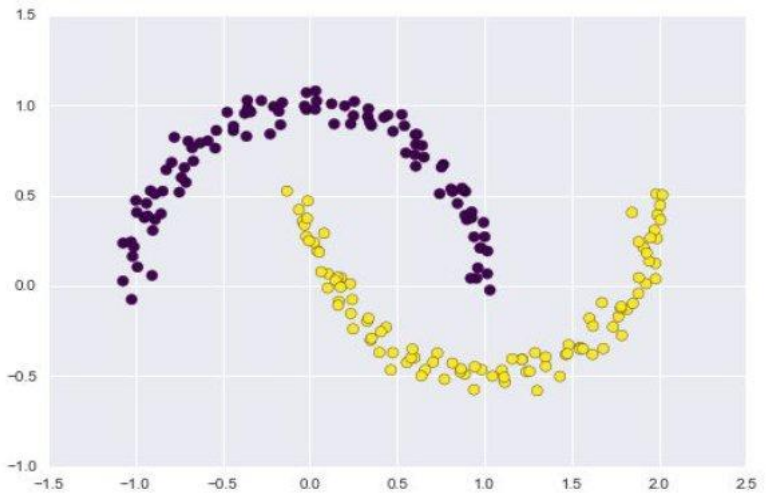
1. Calculate clusters - for each point assign them to closests center.
2. Re-calculate the centers - as the average of all points in a cluster.

The algorithm will switch between these 2 operations until there is a convergence and the cluster centers are stable [proof exists for convergence].

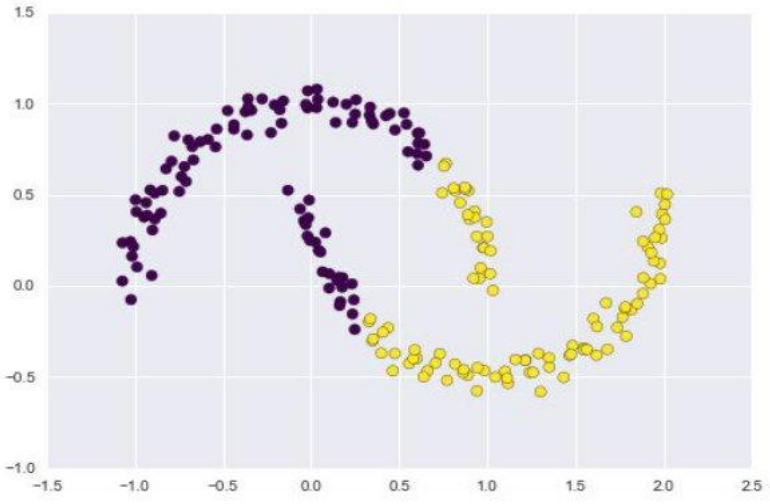
Machine Learning – Clustering.

K-means is linear - fails on non-linear types of data

True cluster



Predicted using k-means



SciKit Learn has support for non-linear clustering, called spectral clustering.

Machine Learning – Clustering.

First:

K-means python tutorial.

Take a look at the
tutorial1kmeans.py

Are predictions for classes as expected?
Try a couple of points.

Machine Learning – Clustering.

First:

K-means python tutorial.

Take a look at the
tutorial1kmeans.py

Are predictions for classes as expected?
Try a couple of points.

Then:

Take a look at exercises 1-2.

Files:
kmeans.py &
moons_dataset_kmeans.py

Decision Trees

Decision Trees

- Useful for : Classification (supervised learning with labels) and Regression (predicting numerical values).

- Works on : multidimensional labelled data (not just 2 dimensions).

- Calculates a Decision Tree that can then be used to predict the classification of new data without labels.

- Interestingly, as many algorithms, seem to produce ”black-box” results, then the output of a decision tree is very easily understood and visualized.

Machine Learning – Decision Trees.

An example : Data used - Iris family of flowers

Iris Setosa



Iris Versicolor



Iris Verginica



Machine Learning – Decision Trees.

Decision Tree - Example - The data - 2D example

3 classes in this dataset:

Red: Setosa

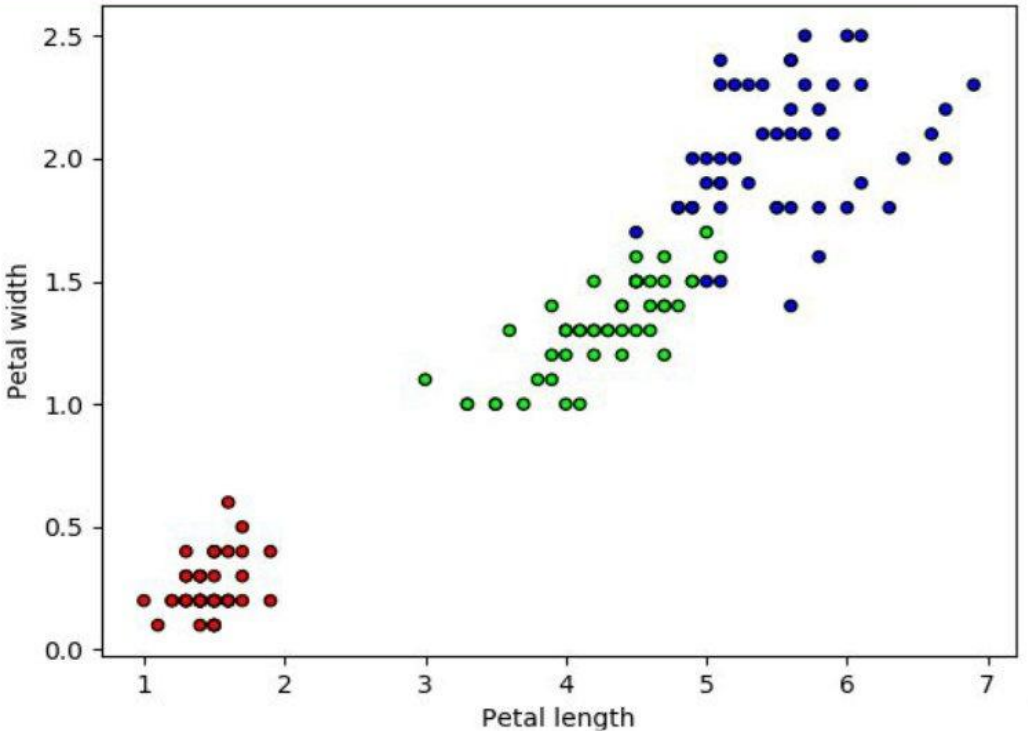
Green: Versicolor

Blue: Verginica

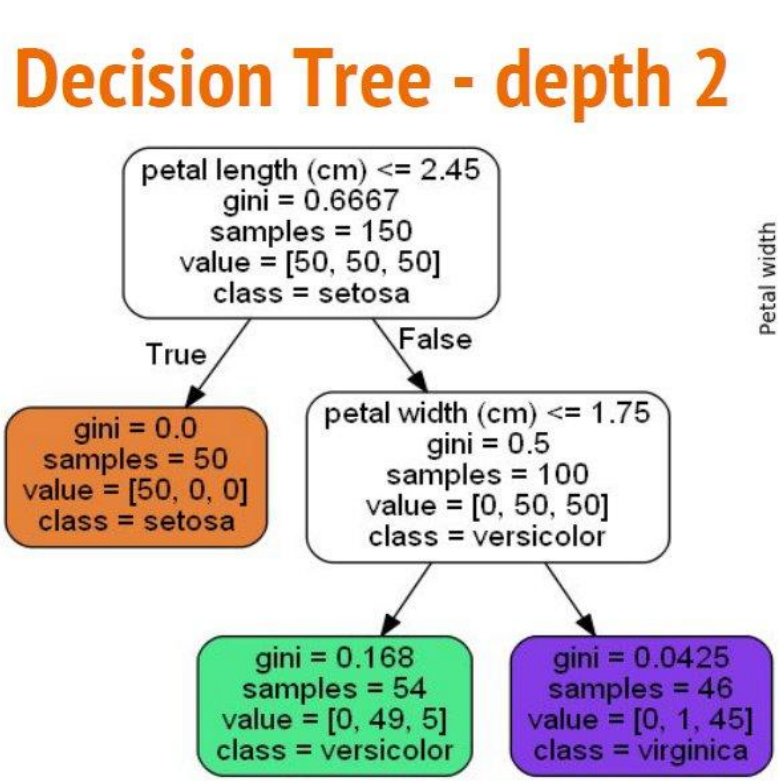
Here they are plotted

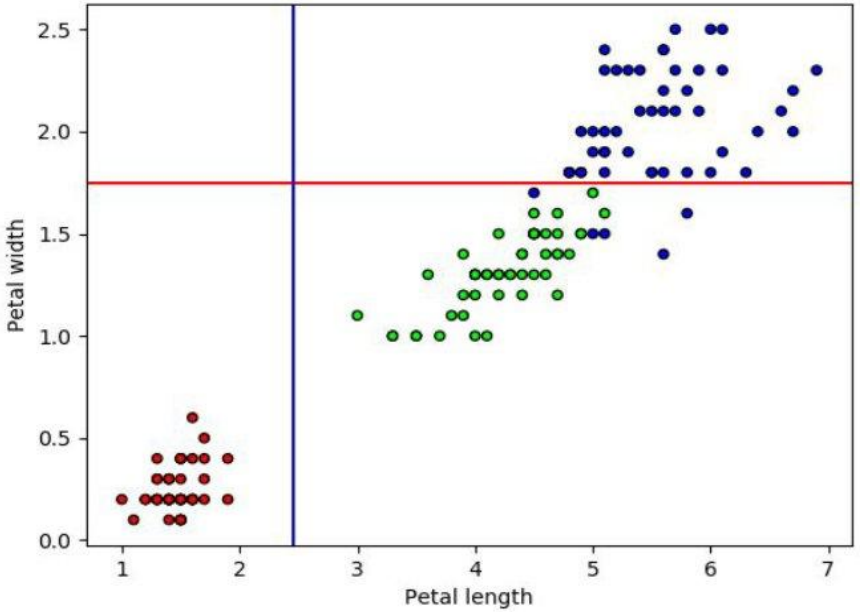
with their labels (colors)

Petal: flower leaves (unit cm)



Machine Learning – Decision Trees.



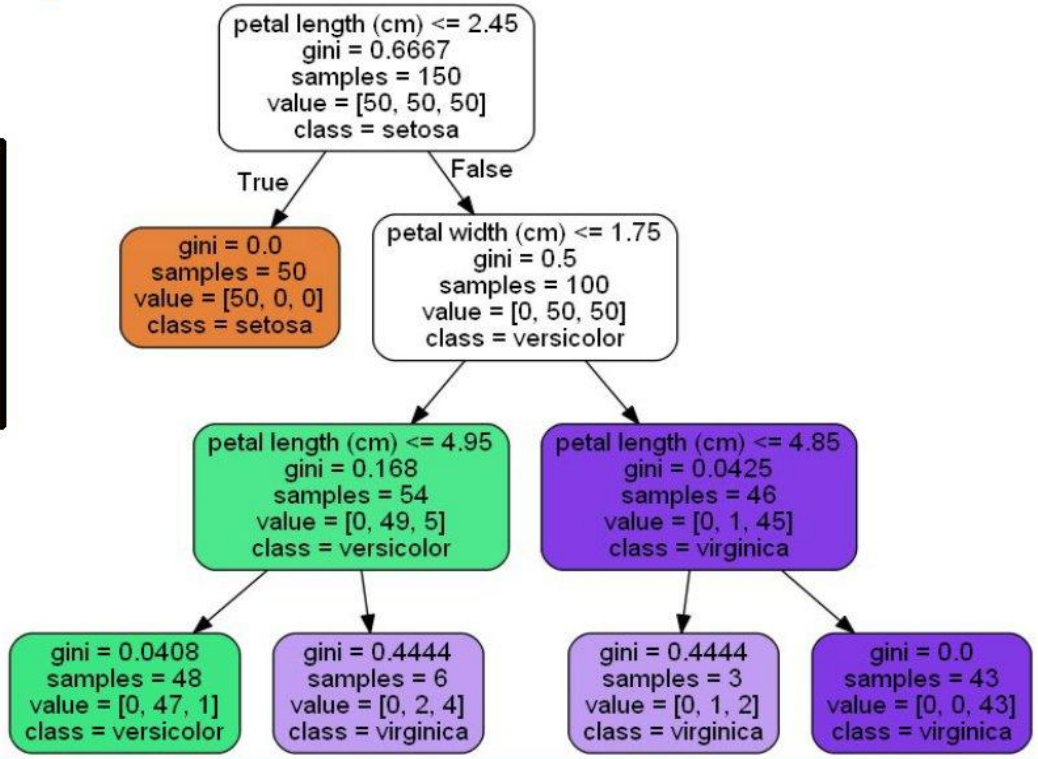


In total 6 points falls in the wrong group, using this tree. [The 5 + 1 in the lowest two boxes].

Machine Learning – Decision Trees.

Decision Tree - depth 3

In total 4 points falls into the wrong category with this tree (0 +1 + 2 + 1 + 0 = 4). So better classification rate, but slower classification



Decision Tree: Training algorithm.

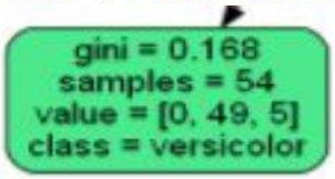
- SciKit-Learn uses the CART training algorithm (other algorithms such as ID3 exists).
- CART always produces binary decision trees. Other algorithms can create n-trees. But CART is very fast.
- Goal of algorithm is to minimize the gini (impurity) at the nodes.

Gini impurity $= 1 - \sum_{i=1}^{J} {p_i}^2$ **(low gini good, high gini bad!)**

Where J is number of classes. Pi is fraction of the items labelled with class i.

If gini = 0, then the node is pure (no errors).

$$\text{Gini} = 1 - (0^2 + (49/54)^2 + (5/54)^2) = 0.168$$



Decision Trees: Strengths and weaknesses.

- Makes no assumptions on the data, so can fit to anything.
- Works on any multi-dimensional data.
- Can lead to overfitting if not careful (but general problem).
- Can be sensitive to small changes in the training data

Decision Tree example:

Run the Decision Tree tutorial
(tutorial2decisiontree.py)

- Try increasing the max height of the tree…

- Discuss in groups (2-3):
  Verify/make sure that you agree with probablity scores for values:
  (2,1), (4,1) & 5,2)

Decision Tree exercise.

Exercise 3.

Get started in groups of 2-3 and discuss how to proceed.

(Then a little later today):
Look in the file moons_dataset_decisiontree.py for much of the needed code, but try to do as much as possible yourself.

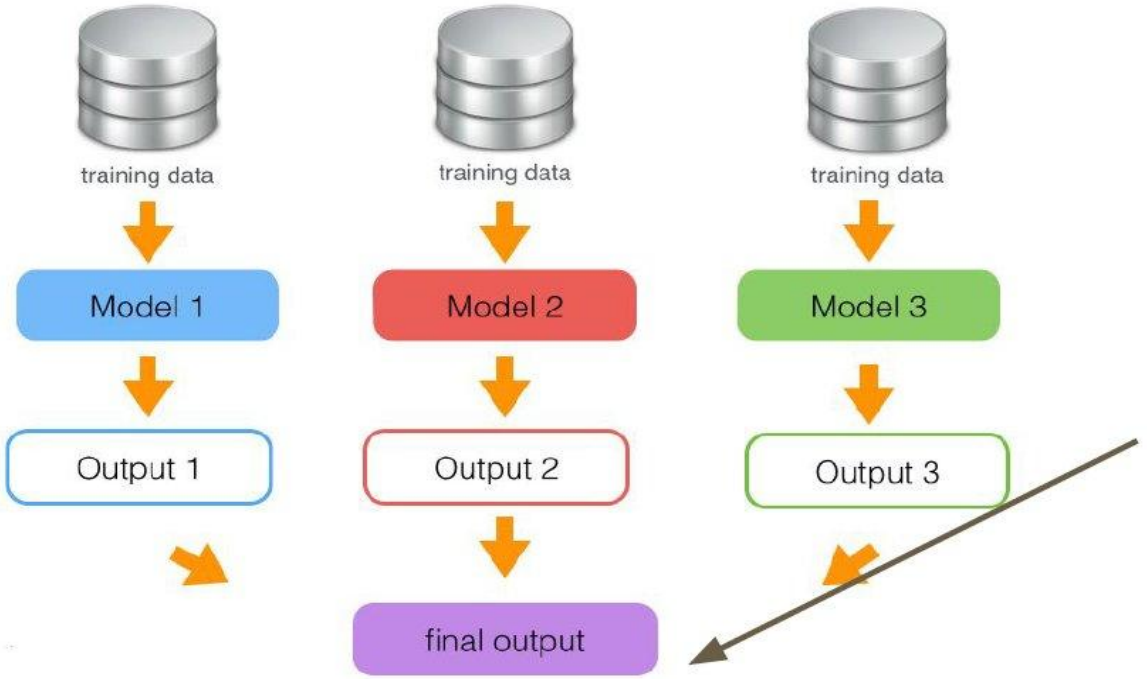
Random Forest & Ensemble methods.

Ensemble methods.

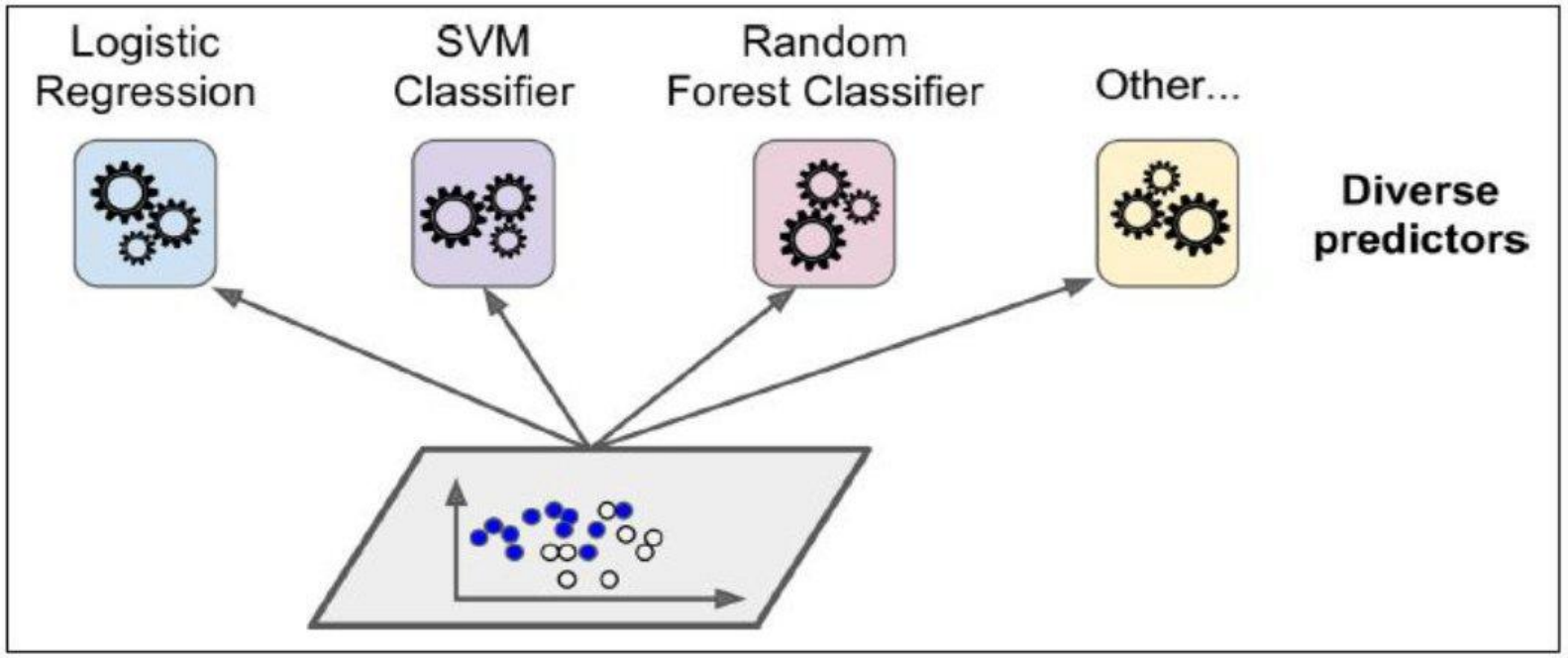


- "Wisdom of the crowds"
- Training a group of predictors.
- This is called "ensemble learning" - ensemble meaning a group of something (ensemble - a group of musicians fx).
- You can then let the predictors each make a prediction and then you can take the "majority vote" or other method of determining the final output.
- A majority vote is also called hard voting.
- Soft voting instead calculates the average probabilities of each prediction across all classifiers and then selects the class with the highest probability. An analogy is in elections: Typical many US states have hard voting, so if you get 51% you win the entire state. Soft voting you get a proportional numbers of mandates (we have soft voting in Denmark).

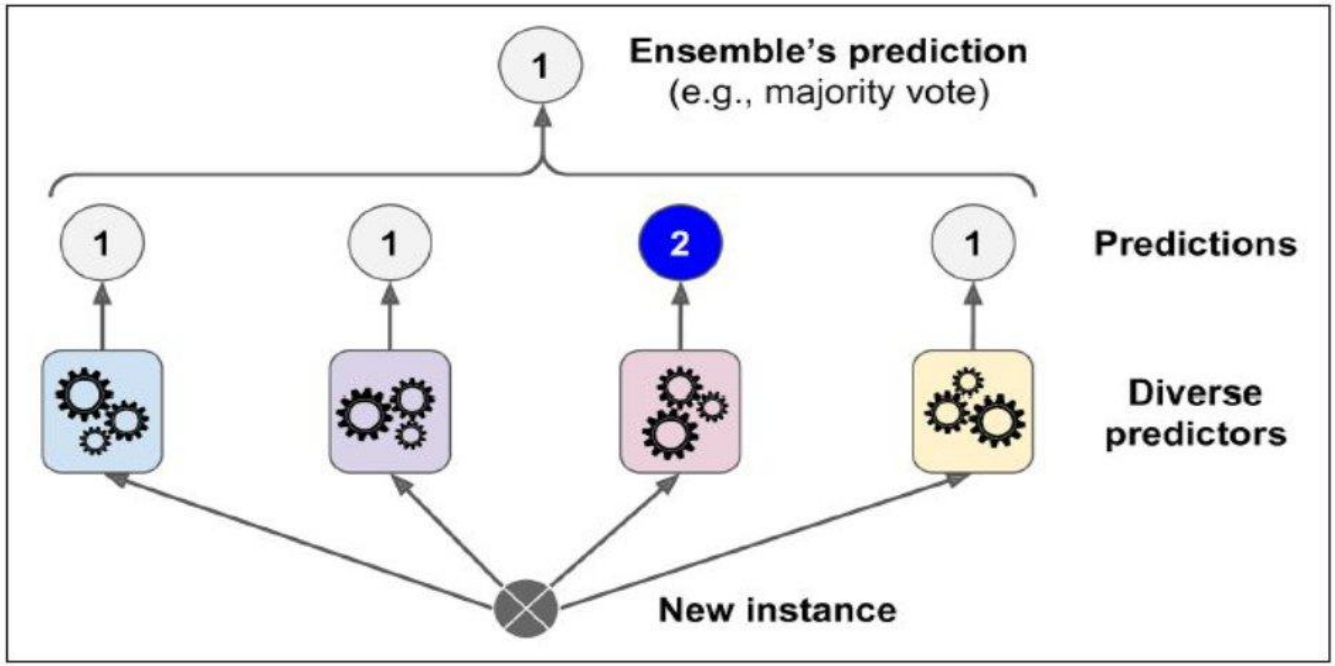
Ensemble learning – overview.



Ensemble learning - training example.



Ensemble Learning - prediction example.



Real case: Netflix

Competition by Netflix to improve movie recommendations in 2009.
Prize: Winners got 1.000.000 $.

Dataset : 100 million movie ratings by anonymous users
Winning method: Used 24 different predictors and specialized ensemble
learning (Gradient Boosted Decision Trees).
https://netflixprize.com/

Research Paper (advanced!):
https://netflixprize.com/assets/GrandPrize2009_BPC_BellKor.pdf

Machine Learning – Random Forest and Ensemble Methods.

Why and when does ensemble learning work?

- Usually used after you have one or more good classifiers already that you want to combine.

- An analogy to why it works: Jeopardy Quiz Show - ask the audience lifeline - usually good try to go with the majority answer (if people are not clones)

Argument from the book:
1000 predictors, each only right in 51% of the cases.
Combine them and take majority vote, you will be right in almost 75 % of the cases (according to the book)

Math proof relates to the law of large numbers. See book, or:

Binomial Probability Calculator: https://www.gigacalculator.com/calculators/binomial-probability-calculator.php

Ensemble learning - creating diversity.

- It's no use to train the same algorithm 10 times on the same training data and then take a vote……
- Generally, 2 approaches can be used:

1. Train **different algorithms** on the **same data** and get a majority vote
2. Train **the same algorithm** on a **different subset** of the training data.

The second approach can be done in two different ways, either you can choose a subset of the train data **without putting the data points back** for other subsets - **called pasting** (so no data is shared between training).
Or you can **put the data points back for other subsets to include these data** points - this gives more training data, but overlap - **called bagging.**

Machine Learning – Random Forest and Ensemble Methods.

Random Forest -

an example of Ensemble learning.

- One of the **most popular ML algorithms used**.
- In Python you can give an argument to the algorithm - (n_estimators=..), where you specify how many trees should be used.
- The trees are then trained on **subsets of the data** (**using bagging**), and then when it comes to prediction a **probabilistic vote** (**soft voting**) takes place amongst the trees (in SciKit-Learn the algorithm is implemented in this way, you could also implement it with hard voting if you wanted).
- **In general, up-to a certain point, the classification accuracy improves with the number of classifiers.**

https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html

https://stats.stackexchange.com/questions/127077/random-forest-probabilistic-prediction-vs-majority-vote

Random Forest and Ensemble Methods example:

Run/ Take a look at tutorial
tutorial3randomforest.py

In groups – Explain to each other what is going on.

Default values, for random forest, are usually pretty good, but there is a lot you can potentially adjust:
https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html

Random Forest and Ensemble Methods exercise:

First:

Run/ Take a look at tutorial tutorial3randomforest.py

In groups – Explain to each other what is going on.

Then:

Exercise 4.

// Use tutorial files for inspiration. //

Random Forest and Ensemble Methods exercise:

First:

Run/ Take a look at tutorial tutorial3randomforest.py

In groups – Explain to each other what is going on.

Then:

Exercise 4.

// Use tutorial files for inspiration. //

Finally:

Exercise 5-6.

Homework

Week 4.
Topics : Clustering (k-means), Decision Trees, Random Forest and Ensemble Learning.

Litterature:
[Hands on Machine Learning] chapter 6 and chapter 7
(**p. 215- 233 ,** p. 199 – 211 v2., look through,
but don't worry too much about the details).

Video:
https://www.youtube.com/watch?v=LDRbO9a6XPU&list=WL&index=3

Exercises.
Do all the exercises for this week.